Identifying Positional Archetypes in the National Women's Soccer League (NWSL) through Clustering

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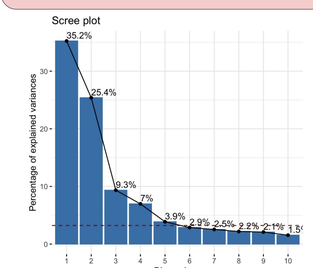
Background

- The National Women's Soccer League is the highest level of professional women's soccer in the United States, consisted of 14 teams in 2024, has now expanded to 16 teams
- Our goals are:
  - **To determine which statistics separate attackers, midfielders and defenders**
  - **To quantify the role defensive midfielders play offensively and defensively**
- Technical analysis in the NWSL has been practically non-existent, worked to utilize publicly available data from the NWSLR repository

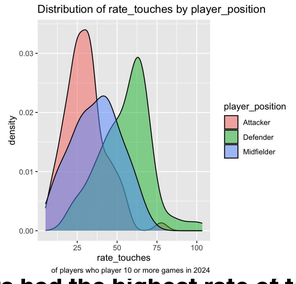
Methods



- Principal Component Analysis
  - To understand which variables contributed to the most variation between players
- Hierarchical Clustering using k = 7 based on 7 types of outfield positions
  - Used complete linkage: $max\{d(x,y):x\in A, y\in B | A, B \in N\}$
- Cluster Standardization:
  - Used $(centroid\ value - variable\ mean)/(variable\ standard\ deviation)$ to standardize each variable for comparison of movement across clusters

Data

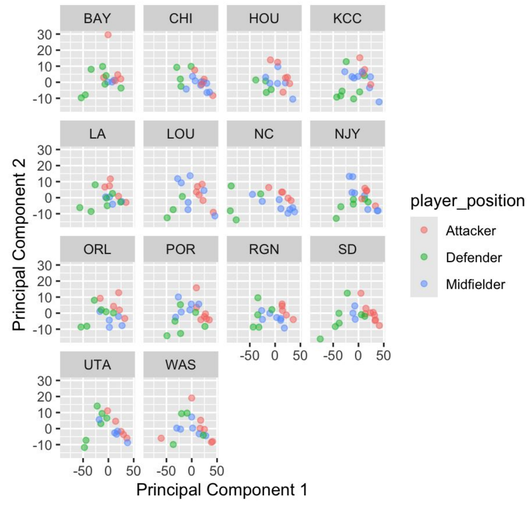
- Used publicly available data gathered by Dror and Jackson [1], accessed via the nwslR package [2]
- Only data from the 2024 regular season and playoffs was used; all Challenge Cup (in-season tournament) data was exclude
- Data was filtered to only include players who played 10+ games, excluding goalkeepers



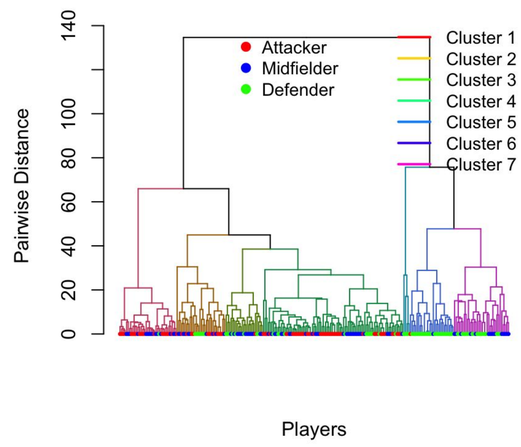
Defenders had the highest rate of touches followed by midfielders then attackers

Offensive Statistics	Defensive Statistics
Key Passes	Recoveries
Through Balls	Interceptions
Touches	Tackled
Crosses Open Play Successful	Tackles Won
Successful Passes Total	Tackles Lost
Total Fouls Won	Tackles Total
Passes Forward	Duels Won
Passes Backwards	Duels Lost
Loss of Possession Total	Duels Ground
Short Passes Successful	Aerial Duels Won
Short Passes Unsuccessful	Aerial Duels Lost
Shots Total	Ground Duels Won
Shots on Target	Duels Ground Lost
Long Passes Successful	
Long Passes Unsuccessful	

Analysis and Results

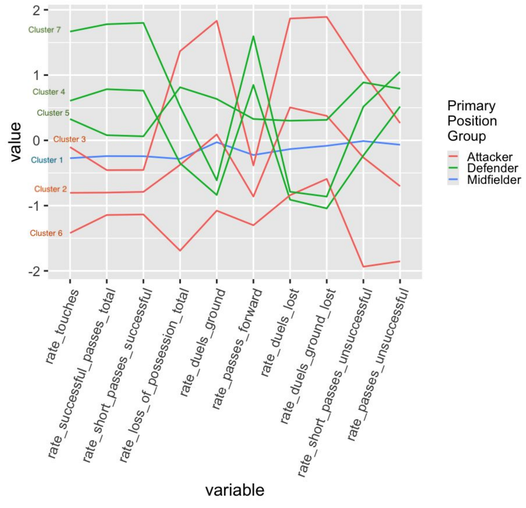


PCA projection of all teams across the league reveals similar distributions of player types except for the North Carolina Courage (NCC).



Clustering reveals defenders are statistically different from attackers and midfielders

Observe one predominantly midfielder cluster, defenders have a higher rate of touches, while midfielders are mostly in between attackers and defenders across all statistics



Conclusion

- Players did not cluster with those who had the same outfield position label
- Defenders and attackers are very statistically different
  - Defenders contributed a higher number of touches, and successful passes (overall, shorts, and forward).
- Many midfielders are either statistically similar to attackers or defenders (defensive midfielders).

References

[1] Dror, Arielle and Jackson, Lydia (2019). nwslData. nwslR/nwslData. Retrieved 2025, from https://github.com/nwslR/nwslData[2] Dror, Arielle and Jackson, Lydia (2019) [Computer software]. nwslR. nwslR. Retrieved 2025,
from https://github.com/nwslR/nwslData
[3] Murtagh, Fionn, and Pedro Contreras. "Algorithms for hierarchical clustering: an overview." Wiley interdisciplinary reviews: data mining and knowledge discovery 2.1 (2012): 86-97.
[4] Dutta, Rishav, Yurko, Ron, and Ventura, Samuel L. (2020). "Unsupervised Methods for Identifying Pass Coverage Among Defensive Backs with NFL Player Tracking Data". arXiv. https://arxiv.org/abs/1906.11373